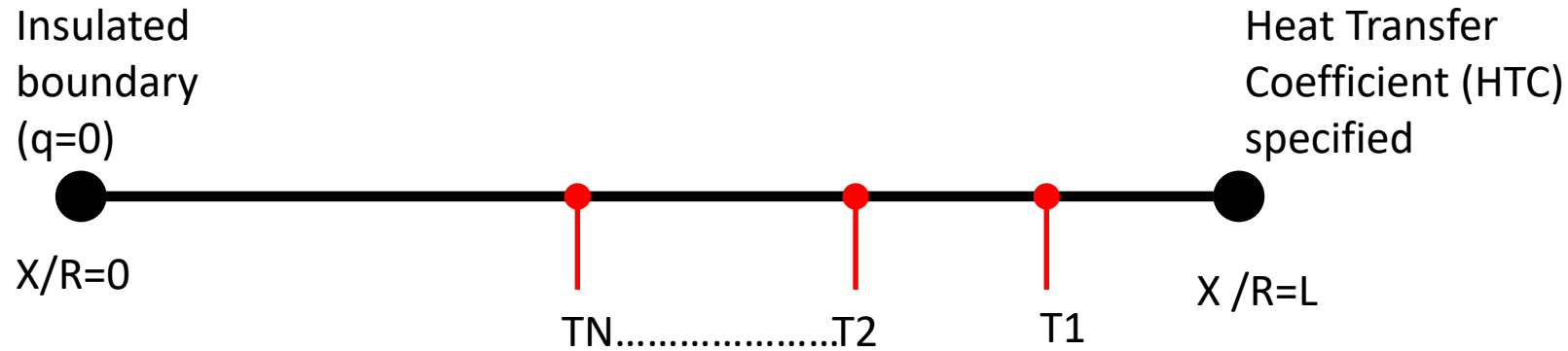


HTC-Based Transient 1D FEM for Cooling Curve Simulation with and without Phase Transformation

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1D Finite Element Model



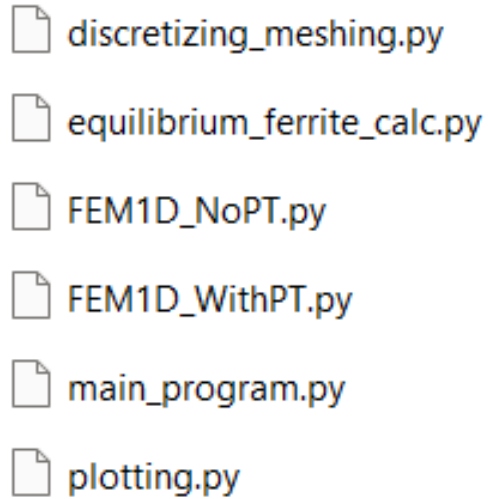
- X or R indicates the X coordinate or the radius depending the type of coordinate system (either cartesian or cylindrical).

Heat conduction equation in Cartesian Coordinate System=> $\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + Q = \rho C \frac{\partial T}{\partial t}$

Heat conduction equation in Cylindrical Coordinate System=> $\frac{\partial}{\partial r} \left(kr \frac{\partial T}{\partial r} \right) + Q = \rho C \frac{\partial T}{\partial t}$

Q is the internal heat generated due to phase transformation. Q =0 if there is no phase transformation.

Download and save all the below mentioned files in a single folder before launching the code.



Python script files:

1. main_program.py => main file to initiate the GUI.
2. discretizing_meshing.py => For meshing and storing geometry details.
3. FEM1D_NoPT.py=> main script file for solution without phase transformation.
4. FEM1D_WithPT.py =>main script file for solution with phase transformation.
5. Equilibrium_ferrite_calc.py => function to calculate the equilibrium ferrite for hypoeutectoid steels
6. Plotting.py => python class for plotting result

How to run the script file?

- The above mentioned python files should be present in the same folder to run the program.

Pre requisites to run the Program:

- Download and install latest version of Python.
- Download Pycharm (community edition) or any other equivalent IDE to view/edit the script.
- Python libraries:
 - Numpy
 - Pandas
 - Scipy
 - Matplotlib

For detailed theoretical understanding of the script, please refer to the following research article:

- Samuel, A., Pranesh Rao, K.M. & Narayan Prabhu, K. Critical Heat Transfer Coefficients for Selection of Quench Media during Heat Treatment of Steels. J. of Materi Eng and Perform 34, 5327–5338 (2025). <https://doi.org/10.1007/s11665-024-09448-w>

How to use?

-Run 'main_program.py' either in command prompt or using any python Integrated Development Environment (IDE) such as pycharm

Graphic User Interface

1D Transient Heat Conduction Equation

Output directory

Output Directory:

Geometry and Mesh details

Coordinate System

☐ Cartesian

☐ Cylindrical

X/R dimension (mm)

X/R division

Solver details

Boundary Heat Transfer Coefficient (W/m^2K)

Initial Temperature (degC)

Ambient Temperature (degC)

time interval, dt (s)

Post Process

Provide X/R locations separated by comma

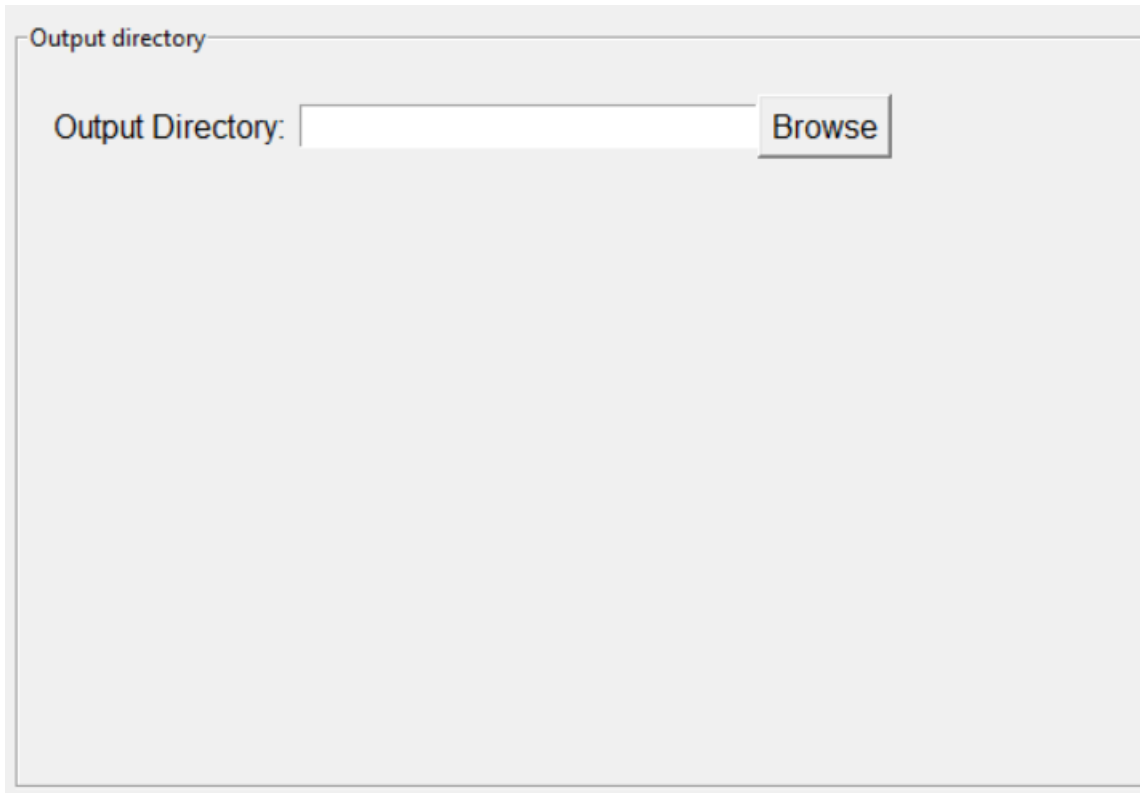
X/R location (mm)

Material details

☐ Without phase transformation

☐ With phase transformation

Section 1: Output Directory



Output directory

Output Directory:

Browse

Step 1: Create a folder in the computer (either in local disk D or E) to store the output from the solver.

Step 2: In the Output directory section of GUI, browse and select the output folder to obtain the folder path.

Section 2: Geometry and Mesh details

Geometry and Mesh details

Coordinate System

☐ Cartesian

☐ Cylindrical

X/R dimension (mm)

X/R division

1. Select the type of coordinate system (either Cartesian or Cylindrical).
2. Provide X dimension or R (radius of cylinder) in mm.
3. Specify the number of divisions in integer (X/R division).
example: if $X=25\text{mm}$ and $\Delta X=0.2\text{mm}$,
then $X \text{ division} = X / \Delta X = 125$.

Note: Press Confirm button after filling all details as it will save all the geometry and mesh details in the output directory.

Section 3: Material Details

Material details

☐ Without phase transformation

☐ With phase transformation

Confirm

Material details

☒ Without phase transformation

☐ With phase transformation

Density (Kg/m^3)

a1

a2

a3

Constant

Thermal Conductivity (W/m K)

a1

a2

a3

Constant

Specific Heat (J/Kg K)

a1

a2

a3

Constant

Confirm

Material Properties can be expressed as a function of temperature as followed.

$$\rho(T) = a_1T^3 + a_2T^2 + a_3T + constant$$
$$k(T) = b_1T^3 + b_2T^2 + b_3T + constant$$
$$C(T) = c_1T^3 + c_2T^2 + c_3T + constant$$

The correlation can be cubic, quadratic, linear or constant.

For cubic: specify all variables (a1, a2, a3, constant).

For quadratic: a1=0, specify a2, a3, and constant.

For linear: a1=0, a2=0, specify a3 and constant.

For constant material property: a1=0,a2=0,a3=0, specify constant.

Once after filling all material details, press confirm to save the material data in the output directory.

Section 3: Material Details Continued

Material details

Without phase transformation

With phase transformation

Temperature and Phase Dependent Thermal Properties

Temperature in degree Celsius

Thermal conductivity (W/mK)

Density (Kg/m3)

Specific heat (J/KgK)

Latent heat (J/Kg)

Load TTT Diagram

Browse

Browse

Browse

Browse

Browse

Critical Temperatures in degree Celsius(Put Ae3=0 if no ferrite transformation)

Ae3

Ae1

Bs

Ms

Initial Austenite Phase Fraction (0-1)

Steel Composition in Wt% for equilibrium Ferrite Calculation (Put 0.0 for non existing element)

C

Cr

Mn

Mo

Ni

Si

Confirm

Order of columns in excel files

1. Hypo eutectoid steel

1a.Thermophysical properties (Density, thermal conductivity, specific heat)

Temperature (deg C)	Austenite	Ferrite	Pearlite	Bainite	Martensite

1b. Latent Heat.

Temperature (deg C)	Ferrite	Pearlite	Bainite	Martensite

1c. TTT data

Temperature (deg C)	Ferrite start time (s) (0.1wt%)	Pearlite start time (s) (0.1wt%)	Pearlite Finish time (s) (99.9 wt%)	Bainite start time (s) (0.1wt%)	Bainite Finish time (s) (99.9wt%)

Note: The program uses PCHIP interpolation, therefore the temperature column should be in the increasing order

Section 3: Material Details Continued

Material details

Without phase transformation

With phase transformation

Temperature and Phase Dependent Thermal Properties

Temperature in degree Celsius

Thermal conductivity (W/mK)

Density (Kg/m3)

Specific heat (J/KgK)

Latent heat (J/Kg)

Load TTT Diagram

Browse

Browse

Browse

Browse

Browse

Critical Temperatures in degree Celsius(Put Ae3=0 if no ferrite transformation)

Ae3

Ae1

Bs

Ms

Initial Austenite Phase Fraction (0-1)

Steel Composition in Wt% for equilibrium Ferrite Calculation (Put 0.0 for non existing element)

C

Cr

Mn

Mo

Ni

Si

Confirm

Order of columns in excel files

2. Hyper eutectoid steel

2a.Thermophysical properties (Density, thermal conductivity, specific heat)

Temperature (deg C)	Austenite	Pearlite	Bainite	Martensite

2b. Latent Heat.

Temperature (deg C)	Pearlite	Bainite	Martensite

2c. TTT data

Temperature (deg C)	Pearlite start time (s) (0.1wt%)	Pearlite Finish time (s) (99.9 wt%)	Bainite start time (s) (0.1wt%)	Bainite Finish time (s) (99.9wt%)

Note: The program uses PCHIP interpolation, therefore the temperature column should be in the increasing order

Section 3: Material Details Continued

Material details

☐ Without phase transformation
☒ With phase transformation

Temperature and Phase Dependent Thermal Properties

Temperature in degree Celsius

Thermal conductivity (W/mK)		Browse
Density (Kg/m3)		Browse
Specific heat (J/KgK)		Browse
Latent heat (J/Kg)		Browse
Load TTT Diagram		Browse

Critical Temperatures in degree Celsius(Put Ae3=0 if no ferrite transformation)

Ae3	
Ae1	
Bs	
Ms	
Initial Austenite Phase Fraction (0-1)	

Steel Composition in Wt% for equilibrium Ferrite Calculation (Put 0.0 for non existing element)

C	
Cr	
Mn	
Mo	
Ni	
Si	

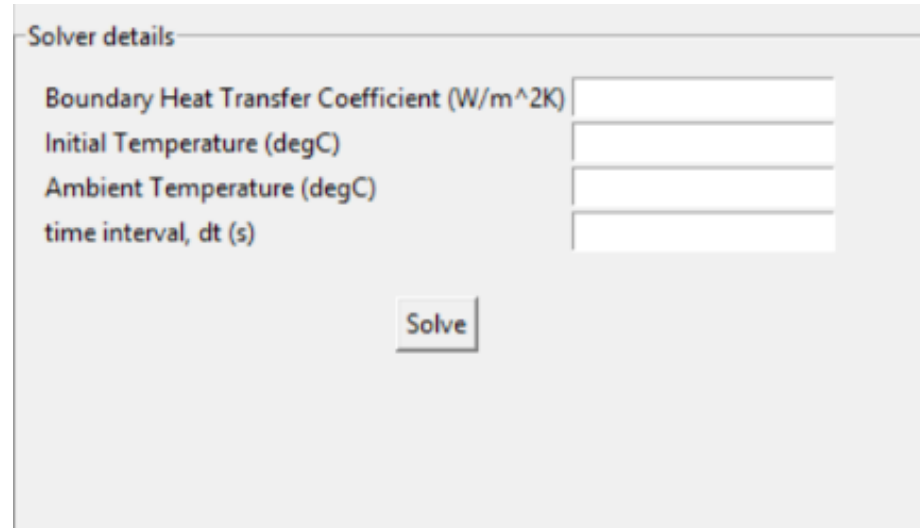
Confirm

Critical Transformation Temperatures

- Enter the critical temperatures in the specified order:
Ae3 → Ae1 → Bs → Ms.
- For **hypereutectoid steel**, set **Ae3 = 0** according to the model convention.
- Ae3= upper eutectoid temperature
- Ae1= lower eutectoid temperature
- Bs = Bainite Start temperature
- Ms= Martensite start temperature
- Initial Austenite Fraction**
Enter the initial austenite fraction.
For a fully austenitized material, use: **1.0**
- Steel Composition**
 - Provide Steel composition in wt% for equilibrium ferrite calculation.
 - Do not leave any cells blank, if a specific element is not in the composition, then mention as 0 in the corresponding text field.
 - For hypereutectoid steel, equilibrium ferrite calculation is not applicable, hence enter 0 in all text fields.

Note: After entering all details, click on confirm to save the material details as text files in the prescribed output directory

Section 4: Solver Details



The image shows a software interface titled "Solver details". It contains four input fields for numerical values: "Boundary Heat Transfer Coefficient (W/m^2K)", "Initial Temperature (degC)", "Ambient Temperature (degC)", and "time interval, dt (s)". Each field is represented by a white rectangular box. Below these fields is a button labeled "Solve".

Solver Details

1. Boundary Heat Transfer Coefficient ($\text{W}/\text{m}^2\cdot\text{K}$)

- Enter the **Heat Transfer Coefficient (HTC)** at the cooling boundary.
- Example: 1000 means **50 $\text{W}/\text{m}^2\cdot\text{K}$** .

2. Initial Temperature ($^{\circ}\text{C}$)

- Enter the temperature of the steel at the beginning of the simulation.
- This is the initial temperature assigned to all FEM nodes.
- Example: 900 $^{\circ}\text{C}$.

3. Ambient Temperature ($^{\circ}\text{C}$)

- Enter the temperature of the surrounding cooling medium/environment.
- The material cools toward this temperature through the HTC boundary condition.
- Example: 30 $^{\circ}\text{C}$.

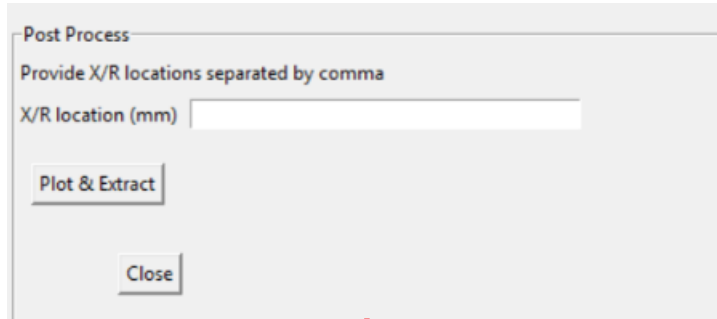
4. Time Interval, dt (s)

- Enter the **time step** used for the transient FEM calculation.
- The solver advances the simulation by this amount of time at every time step.
- Example: 0.1 s.
- Smaller dt $\rightarrow$ finer temporal resolution but generally longer computational time.
- Larger dt $\rightarrow$ faster computation but potentially less accurate resolution of rapid thermal/phase-transformation changes.

5. Solve

- Clicking **Solve** starts the transient FEM calculation using the material, mesh, and solver parameters provided.
- The simulation continues until all nodes reach the prescribed ambient temperature.

Section 5: Post Process



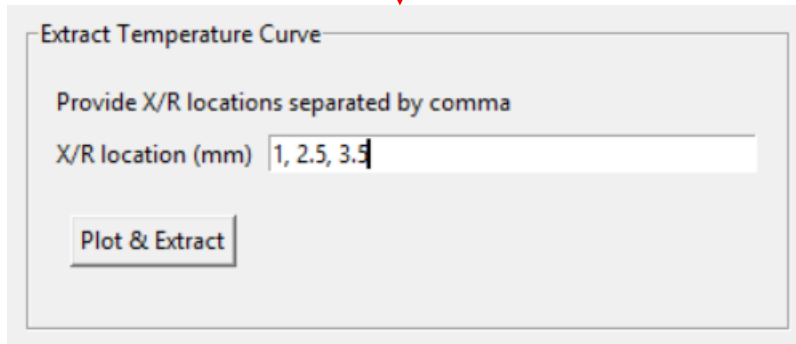
Post Process

Provide X/R locations separated by comma

X/R location (mm)

Plot & Extract

Close



Extract Temperature Curve

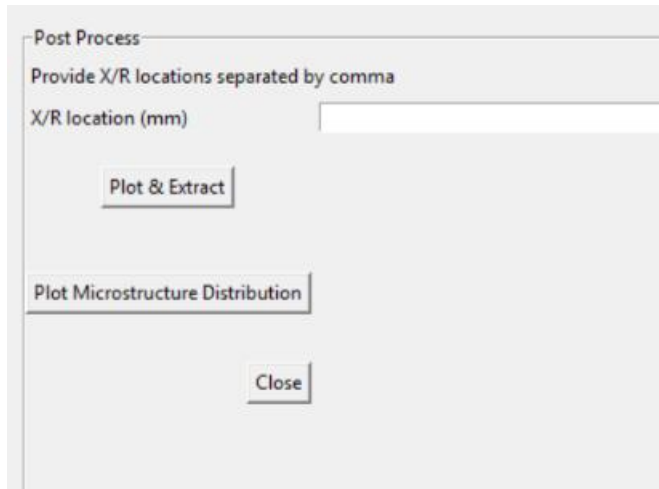
Provide X/R locations separated by comma

X/R location (mm)

Plot & Extract

Option to plot and extract Temperature Vs. Time Curve at any node in the 1D model.

- Provide X or R location in the text field in millimeters to extract the temperature curves.
- Single or multiple temperature curves can be extracted. Multiple location coordinates can be provided by separating with a comma.
- Click Plot and Extract.
- The extracted temperature curves will be available in the output directory with the file name **“Temperature Curve Extract.csv”**.



Post Process

Provide X/R locations separated by comma

X/R location (mm)

Plot & Extract

Plot Microstructure Distribution

Close

Plot Microstructure distribution is activated only when ‘with phase transformation’ option is checked in the ‘Material Details’ section.

Close button will completely terminate the program